

Qingyu LIU

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EDUCATION

Johns Hopkins University Baltimore, MD <i>M.S.E. in Electrical and Computer Engineering</i>	Aug. 2025 – Present
Shanghai Normal University Shanghai, CHN <i>B.S. in Electronic Information Engineering</i> (Outstanding Graduate) GPA: 3.95/4.00 (Rank: 2/523, Top 0.5%)	Oct. 2020 – Jun. 2024

RESEARCH EXPERIENCE

Cross-Lingual F5-TTS v1, Shanghai Jiao Tong University X-LANCE Lab <i>Core Member</i> Supervisor: Prof. Xie Chen	Nov. 2025 – Present
- Proposed synthetic audio prompt generation to replace forced alignment for transcript-free voice cloning training in original Cross-Lingual F5-TTS. - Developed a data construction pipeline via F5-TTS inference for each training sample with quality filtering. - Achieved SOTA performance on the LibriSpeech-PC test-clean benchmark by fine-tuning the pretrained Cross-Lingual F5-TTS with the combination of real and synthetic data.	
Multilingual F5-TTS, Shanghai Jiao Tong University X-LANCE Lab <i>Core Member</i> Supervisor: Prof. Xie Chen	Oct. 2025 – Present
- Developing a multilingual text-to-speech system based on F5-TTS. - Optimizing text tokenizers for multilingual systems to enhance text-speech alignment across languages.	
Cross-Lingual F5-TTS v0, Shanghai Jiao Tong University X-LANCE Lab <i>Core Member</i> Supervisor: Prof. Xie Chen	Mar. 2025 – Sep. 2025
- Developed a novel framework based on F5-TTS with MMS forced alignment that enables cross-lingual zero-shot voice cloning without requiring audio prompt transcripts. - Designed and trained speaking rate predictors to derive speech duration from audio prompt pace. - Deployed an interactive online demo on Hugging Face Space for public evaluation.	
Infant Cry and Snoring Detection, Shanghai Normal University <i>Leading Researcher</i> Supervisor: Prof. Yanhua Long	Nov. 2023 – Jul. 2024
- Constructed and open-sourced the Infant Cry and Snoring Detection (ICSD) dataset for public research use. - Proposed the Competitive CRNN+BEATs method, significantly improving system performance on the ICSD dataset. - Released the ICSD dataset and baseline codes on GitHub.	

INTERNSHIP EXPERIENCE

Johns Hopkins University, Center for Speech and Language Processing <i>Research Assistant</i> Supervisor: Prof. Berrak Sisman	Dec. 2025 – Present
Shanghai Jiao Tong University, X-LANCE Lab <i>Remote Research Assistant</i> Supervisor: Prof. Xie Chen	Aug. 2024 – Present

PUBLICATIONS

- Qingyu Liu**, Yushen Chen, Zhikang Niu, Chunhui Wang, Yunting Yang, Bowen Zhang, Jian Zhao, Pengcheng Zhu, Kai Yu, Xie Chen*. "Cross-Lingual F5-TTS: Towards Language-Agnostic Voice Cloning and Speech Synthesis," *arXiv preprint*.
- Qingyu Liu**, Longfei Song, Dongxing Xu, Yanhua Long* "ICSD: An Open-source Dataset for Infant Cry and Snoring Detection," *Journal of the Audio Engineering Society*.
- Qingyu Liu**, Li Li, Yifan Zhou, Dongxing Xu, Yanhua Long* "A Comparative Study of Deep Learning Methods for Infant Cry Detection," in *NCMMSC*, 2023.

AWARDS

Gold Award , 10th China International College Students "Internet+" Competition	June. 2024
Gold Award , 14th Challenge Cup	July. 2024
First Prize , 14th China National Undergraduate "Innovation, Creativity, and Entrepreneurship" Challenge	May. 2024
Gold Award , 9th "Think Youth" Digital Creation, Innovation & Entrepreneurship Competition	Apr. 2024
Excellent Project Completion , National Undergrad Training Program (Top 5%)	Nov. 2024